DD: FUL-05 — Termination Fulfillment (Orchestration)

Developer implementation guide: `DD_FUL-05-Termination_Fulfillment-DEV_IMPL_GUIDE-v1.0.md`.

1. Purpose

This rewritten DD is the developer-facing version of the module contract `DD_FUL-05-Termination_Fulfillment-v1.0.md`.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0 Orchestrates the technical revocation of a customer's services at the end of the termination flow. Called synchronously by SUB-WF-TERMINATE-01 (per its `callFulfillment` step, R-SUB-WF-TERMINATE-01-WF-1) after billing has accepted the termination intent. FUL-05 loads the Package's constituent Services (per SIP-01), iterates each Service, resolves the target network node + port via the customer's HomePass `service_management_endpoints` map (per RLM-CFG-01), and dispatches to the Provisioner adapter registered for the Service's `provisionerKey` (per PLM-CFG-01) using the `revokeService` interface method. Aggregates outcomes, supports compensation on partial failure, and returns a structured termination outcome. **Aligned design**: this DD mirrors the orchestration model already specified in FUL-03 v1.0 and FUL-04 v1.0. Same Service-loading + Provisioner-dispatch + ledger + compensation pattern, applied to the termination verb. The Provisioner interface method invoked (`revokeService`) is a companion to FUL-03's `activate` / `deactivate` and FUL-04's `applyRestriction` / `liftRestriction` / `suspendService` / `resumeService`; the interface extension is captured as an open coordination item with the PLM team for PLM-CFG-01 v0.9.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	`/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/06_fulfillment/DD_FUL-05-Termination_Fulfillment-v1.0.md`
Version	`v1.0`
DD type	`module contract`
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- `GET /api/termination-fulfillment/tfl_01HZ_KAMAU_MOVED`
- `GET /api/termination-fulfillment?operator=WIK&callerWorkflowRefId=terminate_01HZ_KAMAU_MOVED`

- `GET /api/termination-fulfillment?operator=WIK&reasonCategory=DUNNING_TERMINATION&equipmentDisposition=WRITTEN_OFF&since=2026-05-01`
- `GET /api/termination-fulfillment?operator=WIK&subscriptionId=sub_01HZ_KAMAU_FIBER`
- `GET /api/termination-fulfillment?operator=WIK&terminalState=FAILED_AND_ROLLED_BACK&since=2026-05-01`
- `POST /api/termination-fulfillment/execute`

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- `noc-team`

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- `SubscriptionTerminated`
- `SubscriptionTerminationRejected`

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0

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Aligned design: this DD mirrors the orchestration model already specified in FUL-03 v1.0 and FUL-04 v1.0. Same Service-loading + Provisioner-dispatch + ledger + compensation pattern, applied to the termination verb. The Provisioner interface method invoked (`revokeService`) is a companion to FUL-03's `activate` / `deactivate` and FUL-04's `applyRestriction` / `liftRestriction` / `suspendService` / `resumeService`; the interface extension is captured as an open coordination item with the PLM team for PLM-CFG-01 v0.9.

Glossary

Term	Meaning
Termination	The permanent end of a customer's service relationship. SUB-LM-01 transitions the subscription's `status_code` to `TERMINATED` (terminal). FUL-05 owns the network-plane revoke side of that transition.
Network-plane revoke	The set of operations that turn the customer's network access off: AAA profile disable / delete, OLT port deprovision, vendor profile deletion (Verimatrix shutdown, VoipSwitch revoke). Equipment retrieval is a separate concern owned by OSR-RMA-01 EQP satellite.
`equipmentDisposition`	A discriminator carried in the FUL-05 request envelope, set by SUB-WF-TERMINATE-01 based on the termination context. One of `CUSTOMER_RETAINS` / `PENDING_COLLECTION` / `WRITTEN_OFF`. FUL-05 uses this to decide whether to **delete** the vendor profile (PENDING_COLLECTION, WRITTEN_OFF — equipment will not be reused on this customer) or **disable-only** the profile leaving the auth record in place (CUSTOMER_RETAINS — rare; only when operator policy keeps the AAA record around for legal-retention reasons). FUL-05 does NOT dispatch a pickup tech — that is OSR-RMA-01.
`reasonCategory`	The high-level termination reason group, set by SUB-WF-TERMINATE-01 from its `termination_trigger`: `CUSTOMER_REQUESTED` / `ADMIN_DECISION` / `DUNNING_TERMINATION` / `CONTRACT_EXPIRED`. FUL-05 carries this in the ledger and may pass to the Provisioner for vendor-specific behavior (e.g., a fraud termination may be flagged in AAA logs differently from a customer-requested one).
Service	A row in PLM-CFG-01's Service catalog. Same definition as FUL-03 / FUL-04. FUL-05 operates only on `CONTINUOUS` + `is_addressable = true` Services.

Term	Meaning
Provisioner adapter	The Java bean implementing the `Provisioner` interface (defined in PLM-CFG-01). Same adapters as FUL-03 / FUL-04. FUL-05 calls the `revokeService` method (new in v0.9 of the interface); compensation uses `restoreService` (inverse, used only for rollback within the termination attempt).
Target node code + port	The pair `(nodeCode, port)` resolved from `homepass.service_management_endpoints[service.serviceManagementKey]`. Same resolution as FUL-03 / FUL-04.
Fulfillment attempt	One end-to-end execution of FUL-05 for a given `(subscriptionId, fulfillmentCorrelationId)`. Idempotency-scoped on this tuple. Recorded in `termination_fulfillment_ledger`.
Service outcome	The result of revoking one Service during one attempt: `SUCCESS` / `FAILURE` / `SKIPPED` / `SUCCESS_THEN_COMPENSATED`. Recorded in `termination_fulfillment_service_outcome`.

A — Target

1. What this process is about

SUB-WF-TERMINATE-01 reaches the `callFulfillment` step in its saga, after billing has applied the termination intent (any final charges or refund credits posted). The saga calls FUL-05 synchronously with the canonical envelope. FUL-05 then:

Load:

- The Package (SIP-01) via the subscription's `packageRef`. Read `service_refs[]`.
- For each Service ref, load the Service def (PLM-CFG-01). Read `networkProfileShape.provisionerKey`, `serviceManagementKey`, `consumption_model`, `is_addressable`.
- The HomePass (RLM-CFG-01) via the subscription's `homepassId`. Read `service_management_endpoints`.

Filter: skip Services where `is_addressable = false` (no network state to revoke). Skip Services where `consumption_model != CONTINUOUS` (defensive).

For each remaining Service, in order:

1. Resolve target via `homepass.service_management_endpoints[service.serviceManagementKey]` → `{nodeCode, port}`. If `serviceManagementKey` is null (partner-fulfilled), pass `(null, null)`; the adapter sources target from its own config. Missing endpoint key → `NO_HOMEPASS_ENDPOINT_FOR_KEY`. Null port → `HOMEPASS_PORT_NOT_CAPTURED`.
2. Look up the Provisioner adapter by `service.networkProfileShape.provisionerKey`. Missing adapter → `UNKNOWN_PROVISIONER_KEY`.
3. Invoke `Provisioner.revokeService(reasonCategory, equipmentDisposition, params, customer, targetNodeCode, targetPort)`. The adapter performs the vendor-specific termination: AAA profile delete/disable, OLT port deprovision, vendor profile shutdown. Returns a structured `ProvisionerOutcome` with `success`, `adapterRefs`, `failureReason`, `failureDetail`.
4. Record the Service outcome in `termination_fulfillment_service_outcome`.

Aggregate: After all Services have a recorded outcome (or after the first FAILURE if `partialAllowed = false`), compute the terminal state. Apply compensation if needed.

Audit: Write `termination_fulfillment_ledger` and `termination_fulfillment_service_outcome` rows. Emit `TerminationFulfillmentCompleted` / `TerminationFulfillmentFailed` Kafka events.

Return to caller: Hand control back to SUB-WF-TERMINATE-01. The saga decides what to do with the outcome — on success, the saga commits the subscription to TERMINATED; on failure, the saga reverts to the prior status and emits `SubscriptionTerminationRejected`.

Flow overview — customer-requested termination of a triple-play subscription

```
SUB-WF-TERMINATE-01 reaches callFulfillment after billing intent OK
        ■
        ▼
POST /api/termination-fulfillment/execute
{ subscriptionId, customerId, operatorCode, packageRef,
  homepassId, reasonCategory: "CUSTOMER_REQUESTED",
  equipmentDisposition: "PENDING_COLLECTION",
```

```
  equipmentBindings: [{equipmentRequirement:"GPON_ONT", equipmentRef:"ont_01HZ..."},
                      {equipmentRequirement:"ATA",      equipmentRef:"ata_01HZ..."}],
  fulfillmentCorrelationId, policy? }
        ■
        ▼
FUL-05 loads Package, Services, HomePass, resolves targets,
dispatches Provisioner.revokeService() per Service in order
        ■
        ▼
Aggregate outcomes → write ledger → emit Kafka event → return
{ fulfillmentId, terminalState, services[] }
        ■
        ▼
SUB-WF-TERMINATE-01 commits → subscription TERMINATED, emits SubscriptionTerminated
        ■
        ▼
OSR-RMA-01 EQP satellite picks up the equipment collection (separate flow)
```

Out of scope

- **Physical equipment retrieval / pickup WO dispatch** — owned by OSR-RMA-01 EQP satellite, triggered separately by SUB-WF-TERMINATE-01 or downstream consumers reacting to the `SubscriptionTerminated` event. FUL-05 carries `equipmentDisposition` for the Provisioner's vendor-side behavior (delete vs disable-only) but does NOT call any pickup or work-order system.
- **Equipment instance lifecycle transitions** (e.g., `IN_FIELD_ACTIVE` → `RECOVERED_BY_CONTRACTOR`) — owned by OSR-INSTANCE-01, driven by OSR-RMA-01 workflow events.
- **Subscription state transition to TERMINATED** — owned by SUB-LM-01 via SUB-WF-TERMINATE-01. FUL-05 returns outcome; the saga transitions state.
- **Final invoice / pro-rata refund / Early Termination Fee** — owned by BIL-01 + BIL-02 via the calling workflow's `callBilling` step BEFORE `callFulfillment`. FUL-05 does not call BIL-01.
- **Customer notification of termination** — owned by NOT-01 reacting to the `SubscriptionTerminated` event emitted by SUB-WF-TERMINATE-01 after FUL-05 returns success.
- **Activation / restriction / suspension verbs** — owned by FUL-03 (`activate`, `deactivate`) and FUL-04 (`applyRestriction`, `liftRestriction`, `suspendService`, `resumeService`). FUL-05 only owns `revokeService`.
- **Per-vendor protocol details** — wrapped inside the Provisioner adapter / Gateway code beans (NMSGateway, AAAGateway, VoiceGateway, VerimatrixGateway). FUL-05 only invokes the typed interface.
- **Reversal of a committed termination** — irreversible by design. Once SUB-LM-01 reaches TERMINATED, reinstatement requires a new subscription. The `restoreService` Provisioner method is used ONLY for in-attempt compensation when FUL-05's own chain fails mid-way; it is NOT exposed as a public termination-rollback path.

2. Business rules

Rules grouped: `EN` (entry / envelope), `LO` (loading / resolution), `OR` (ordering), `DI` (dispatch), `AG` (aggregation), `CO` (compensation), `LE` (ledger), `EV` (event emission).

Rule group EN — Entry and envelope

ID	Rule	Triggered on
R-FUL-05-EN-1	FUL-05 has ONE entry point: `POST /api/termination-fulfillment/execute`. Called synchronously by SUB-WF-TERMINATE-01's `callFulfillment` step.	Every invocation
R-FUL-05-EN-2	The envelope MUST carry: `subscriptionId`, `customerId`, `operatorCode`, `packageRef`, `homepassId`, `reasonCategory` (`CUSTOMER_REQUESTED` / `ADMIN_DECISION` / `DUNNING_TERMINATION` / `CONTRACT_EXPIRED`), `equipmentDisposition` (`CUSTOMER_RETAINS` / `PENDING_COLLECTION` / `WRITTEN_OFF`), `equipmentBindings[]` (informational — for the Provisioner's vendor-side context), `fulfillmentCorrelationId`, `callerWorkflowKind`, `callerWorkflowRefId`. Missing any required field → 400 `MISSING_REQUIRED_FIELD`.	Request validation

ID	Rule	Triggered on
R-FUL-05-EN-3	`reasonCategory` values are exactly the four termination triggers defined by SUB-WF-TERMINATE-01 R-WF-R-1. Any other value → 400 `INVALID_REASON_CATEGORY`.	Request validation
R-FUL-05-EN-4	`equipmentDisposition` values are exactly the three states defined by SUB-WF-TERMINATE-01 R-WF-3: `CUSTOMER_RETAINS` / `PENDING_COLLECTION` / `WRITTEN_OFF`. Any other value → 400 `INVALID_EQUIPMENT_DISPOSITION`.	Request validation
R-FUL-05-EN-5	Idempotency on `(subscriptionId, fulfillmentCorrelationId)`. If a fulfillment with the same tuple has already terminated, return the prior outcome unchanged. If a tuple is currently IN_PROGRESS, return `409 FULFILLMENT_IN_PROGRESS`.	Idempotency
R-FUL-05-EN-6	`fulfillmentCorrelationId` is the saga's framework-supplied operation ID (per SUB-WF-FRAMEWORK-01). Same value is reused on saga retries.	Provided by caller
R-FUL-05-EN-7	The optional `policy` envelope defines: `partialAllowed` (default `false` — any Service failure aborts and triggers compensation), `serviceOrder` (default — services processed in the order declared in `Package.serviceRefs[]`), `perServiceTimeoutMs`, `overallTimeoutMs`.	Policy resolution
R-FUL-05-EN-8	Total time budget from request to response: 60 seconds (matching SUB-WF-TERMINATE-01's `FULFILLMENT_TIMEOUT` of 60s per R-WF-4). FUL-05 monitors elapsed time; if `overallTimeoutMs` is reached mid-orchestration, the in-flight Service completes (its own timeout governs) but no further Services are dispatched. Treated as partial failure subject to compensation per policy.	Overall time budget

Rule group LO — Loading and resolution

ID	Rule	Triggered on
R-FUL-05-LO-1	Loads the Package via SIP-01 (read-only) using `packageRef`. Reads `service_refs[]`. If the Package does not exist, the fulfillment fails with `PACKAGE_NOT_FOUND` — without invoking any Provisioner. Note: Package status (Active vs Retired) is NOT a precondition for FUL-05 — terminations can happen on Subscriptions whose Package has since been retired.	Package load
R-FUL-05-LO-2	For each Service ref, loads the Service def via PLM-CFG-01. Reads `consumption_model`, `is_addressable`, `network_profile_shape`, `service_management_key`. Services with `consumption_model != CONTINUOUS` are skipped (`outcome = SKIPPED_NOT_ADDRESSABLE`). Services with `is_addressable = false` are skipped (`outcome = SKIPPED_NOT_ADDRESSABLE`).	Service load
R-FUL-05-LO-3	Loads the HomePass via RLM-CFG-01 (read-only) using `homepassId`. Reads `service_management_endpoints`. HomePass state does NOT block termination — a HomePass in any state can have services revoked from it (the customer is leaving; we want the network state cleaned up regardless of HomePass-level conditions). The chain proceeds; per-Service endpoint resolution may still fail per R-LO-4.	HomePass load

ID	Rule	Triggered on
R-FUL-05-LO-4	For each Service with non-null `service_management_key`, resolves the target via `homepass.service_management_endpoints[service.service_management_key]`. Missing key → `NO_HOMEPASS_ENDPOINT_FOR_KEY` (Service-level failure). Null port → `HOMEPASS_PORT_NOT_CAPTURED` (Service-level failure).	Endpoint resolution
R-FUL-05-LO-5	For each Service with null `service_management_key` (partner-fulfilled), passes `(null, null)` to the Provisioner. The adapter sources target from its own config.	Partner-fulfilled lookup
R-FUL-05-LO-6	Resolves the Provisioner adapter by `service.network_profile_shape.provisionerKey` against the in-memory adapter registry. Missing bean → `UNKNOWN_PROVISIONER_KEY` (Service-level failure).	Provisioner lookup

Rule group OR — Ordering

ID	Rule	Triggered on
R-FUL-05-OR-1	Services dispatched in the order specified in `policy.serviceOrder`, or — if not provided — in the **reverse** of the order declared by `Package.serviceRefs[]`. Reverse ordering is the recommended default for termination (revoke voice and TV before data so the customer doesn't get a brief window of voice-without-data or TV-without-internet that could surface as a support call). FUL-05 reverses automatically when no explicit policy is supplied.	Service dispatch
R-FUL-05-OR-2	`policy.serviceOrder` entries referencing a Service NOT in the Package are silently skipped. Services present in the Package but absent from the policy are appended at the end (defensive).	Defensive
R-FUL-05-OR-3	Services dispatched SEQUENTIALLY (synchronously) for v1.0. Future v1.x may parallelize.	Behavior

Rule group DI — Dispatch

ID	Rule	Triggered on
R-FUL-05-DI-1	Invokes `Provisioner.revokeService(reasonCategory, equipmentDisposition, params, customer, targetNodeCode, targetPort)`. `params` is `service.networkProfileShape.params`. The adapter performs vendor-specific termination based on `equipmentDisposition`: `CUSTOMER_RETAINS` → disable AAA profile (preserve record); `PENDING_COLLECTION` / `WRITTEN_OFF` → delete AAA profile + deprovision OLT port + shutdown vendor service (full clean-out). Returns `ProvisionerOutcome` with `success`, `adapterRefs`, `failureReason`, `failureDetail`.	Per Service
R-FUL-05-DI-2	Per-Service timeout from `policy.perServiceTimeoutMs` (default from `ful_05_config`, typically 25 seconds). On timeout, Service outcome is `FAILURE` / `ADAPTER_TIMEOUT`.	Adapter call
R-FUL-05-DI-3	If the adapter throws (Java exception), Service outcome is `FAILURE` / `ADAPTER_ERROR` with `failure_detail` = exception message.	Adapter error
R-FUL-05-DI-4	The full `ProvisionerOutcome` payload is stored in `termination_fulfillment_service_outcome.provisioner_outcome` as JSONB. Persistent record of vendor-specific identifiers needed for later compensation or troubleshooting.	Per outcome

Rule group AG — Aggregation

ID	Rule	Triggered on
R-FUL-05-AG-1	If `policy.partialAllowed = false` (default): chain aborts on first FAILURE. Subsequent Services get `outcome = SKIPPED_ABORTED_CHAIN`. Terminal state is `FAILED_AND_ROLLED_BACK` if any prior Service succeeded (triggers compensation) or `FAILED_AT_FIRST_SERVICE` if no prior Service succeeded.	First failure
R-FUL-05-AG-2	If `policy.partialAllowed = true`: chain continues through all Services. Terminal state is `FULLY_REVOKED` / `PARTIALLY_REVOKED` / `FAILED_AT_FIRST_SERVICE`. Compensation is NOT invoked — the saga sees the partial outcome and decides. NOTE: for DUNNING_TERMINATION operators may prefer `partialAllowed = true` since partial revocation is still progress toward debt protection.	All Services attempted
R-FUL-05-AG-3	If the Package contained zero `is_addressable` Services after the filter, terminal state is `NOTHING_TO_REVOKE` (no adapter calls, immediate return). Logged for operator review.	Empty Service set

Rule group CO — Compensation

ID	Rule	Triggered on
R-FUL-05-CO-1	Compensation is invoked when terminal state is `FAILED_AND_ROLLED_BACK`. For each previously-successful Service, FUL-05 calls `Provisioner.restoreService(reasonCategory, equipmentDisposition, params, customer, targetNodeCode, targetPort)` to undo the revoke. Order: reverse of original dispatch (so the last-revoked is first-restored).	Compensation trigger
R-FUL-05-CO-2	The `Provisioner.restoreService` method is the `revokeService` inverse for compensation only. It is NOT the path for legitimate reactivation — that is FUL-03 `activate()`. The two are separate to keep the compensation contract clean. The interface extension is captured as an open coordination item for PLM-CFG-01 v0.9 (alongside FUL-04's interface additions).	Interface extension
R-FUL-05-CO-3	Compensation results recorded in `termination_fulfillment_service_outcome.compensation_status`: `COMPENSATED` / `COMPENSATION_FAILED` / `NOT_APPLICABLE`. A failed compensation does NOT abort the overall flow — FUL-05 logs, emits `CompensationFailed` for NOC alerting, continues.	Per compensation result
R-FUL-05-CO-4	After all compensations attempted, FUL-05 writes the final ledger row and emits `TerminationFulfillmentFailed`. Terminal state remains `FAILED_AND_ROLLED_BACK` even if some compensations failed — the operator (NOC) handles cleanup. Subscription stays in its prior status (NOT TERMINATED).	Final emission
R-FUL-05-CO-5	Compensation is NOT invoked for `FAILED_AT_FIRST_SERVICE` (nothing succeeded) or `PARTIALLY_REVOKED` (operator policy says partial is OK).	Skip cases

Rule group LE — Ledger

ID	Rule	Triggered on
R-FUL-05-LE-1	One ledger row per fulfillment attempt in `termination_fulfillment_ledger`. Unique on `(subscription_id, fulfillment_correlation_id)`. Carries the envelope fields, the terminal state, timestamps, and aggregated counts.	Every attempt

ID	Rule	Triggered on
R-FUL-05-LE-2	One service-outcome row per Service in the Package per attempt in `termination_fulfillment_service_outcome`. Carries the Service ref, dispatch order, the `(targetNodeCode, targetPort)` resolved, the outcome, failure reason if any, the full `provisioner_outcome` JSONB, and the compensation status if applicable.	Every dispatched Service
R-FUL-05-LE-3	Both tables are append-only at the application layer. Updates only happen during the same fulfillment attempt (e.g., flipping a Service outcome to `SUCCESS_THEN_COMPENSATED` when compensation runs). After the ledger row reaches a terminal state, no further updates.	Append-only
R-FUL-05-LE-4	Retention: indefinite. Termination is the irreversible end of a customer relationship; the ledger is the permanent record of what happened on the network plane at the moment of revoke. Workshop sources flagged daily BSS↔NMS reconciliation; this ledger participates in that reconciliation.	Retention

Rule group EV — Event emission

ID	Rule	Triggered on
R-FUL-05-EV-1	On any terminal state, emit one Kafka event with the full ledger contents: `TerminationFulfillmentCompleted` (terminal `FULLY_REVOKED`), `TerminationFulfillmentPartial` (terminal `PARTIALLY_REVOKED`), `TerminationFulfillmentFailed` (terminal `FAILED_AND_ROLLED_BACK` / `FAILED_AT_FIRST_SERVICE` / `NOTHING_TO_REVOKE`).	Terminal state reached
R-FUL-05-EV-2	On compensation failure, emit `CompensationFailed` (one per failed compensation) in addition to the terminal event. NOC subscribes to `CompensationFailed` for manual cleanup alerting.	Compensation failure
R-FUL-05-EV-3	All Kafka events carry `fulfillmentId`, `subscriptionId`, `customerId`, `operatorCode`, `reasonCategory`, `equipmentDisposition`, plus the per-service summary. The saga's `fulfillmentCorrelationId` is also carried for cross-system correlation with billing / SUB-LM events.	Event payload

3. Schemas

`termination_fulfillment_ledger`

Column	Type	Notes
`id`	TEXT PK	ULID. Example: `tfl_01HZX3K9M2P7Q4R8T6W1Y3Z5N7`
`operator_code`	TEXT NOT NULL	Values: `WIK` / `WUG` / `WTZ` / `YASSN`
`subscription_id`	TEXT NOT NULL	FK to subscription. Example: `sub_01HX5M2K9P3Q7R4T6W8Y0Z2N4`
`customer_id`	TEXT NOT NULL	Denormalized for fast lookup. Example: `cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6`
`fulfillment_correlation_id`	TEXT NOT NULL	Saga's operation ID. Example: `saga-terminate-sub_01HX-20260518T103000`
`caller_workflow_kind`	TEXT NOT NULL	`SUB_WF_TERMINATE` (v1.0 only caller). Example: `SUB_WF_TERMINATE`
`caller_workflow_ref_id`	TEXT NOT NULL	Calling workflow's saga ID. Example: `terminate_01HZ_KAMAU_MOVED`
`package_ref`	TEXT NOT NULL	Resolved from subscription at fulfillment time. Example: `pkg_INET_VOIP_RES`
`homepass_id`	TEXT NOT NULL	Resolved from subscription at fulfillment time. Example: `hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P`

Column	Type	Notes
reason_category	TEXT NOT NULL	CUSTOMER_REQUESTED / ADMIN_DECISION / DUNNING_TERMINATION / CONTRACT_EXPIRED
equipment_disposition	TEXT NOT NULL	CUSTOMER_RETAINS / PENDING_COLLECTION / WRITTEN_OFF
equipment_bindings	JSONB NOT NULL	Array passed from caller — informational for the Provisioner. Example: [{"equipmentRequirement":"GPON_ONT","equipmentRef":"ont_01HZ..."}]
policy_partial_allowed	BOOLEAN NOT NULL	Effective policy. Example: false
policy_service_order	JSONB NULL	Resolved order if explicit. Example: ["svc_VOIP_LINE", "svc_INET_100M"]
terminal_state	TEXT NOT NULL	FULLY_REVOKED / PARTIALLY_REVOKED / FAILED_AT_FIRST_SERVICE / FAILED_AND_ROLLED_BACK / NOTHING_TO_REVOKE / IN_PROGRESS
services_dispatched_count	INTEGER NOT NULL	Number of Services for which a Provisioner method was invoked. Example: 2
services_succeeded_count	INTEGER NOT NULL	Example: 2
services_failed_count	INTEGER NOT NULL	Example: 0
services_skipped_count	INTEGER NOT NULL	Example: 0
services_compensated_count	INTEGER NOT NULL	Example: 0
started_at	TIMESTAMP NOT NULL	Example: 2026-05-18 10:30:00.123
completed_at	TIMESTAMP NULL	Null while IN_PROGRESS. Example: 2026-05-18 10:30:18.872
duration_ms	INTEGER NULL	Example: 18749

Indexes:

- UNIQUE on (subscription_id, fulfillment_correlation_id) — enforces R-EN-5 idempotency
- idx_tfl_subscription on (subscription_id, started_at DESC) — recent fulfillments per Subscription
- idx_tfl_terminal_state on (operator_code, terminal_state, started_at DESC) — ops dashboards
- idx_tfl_caller on (caller_workflow_kind, caller_workflow_ref_id) — trace back from workflow
- idx_tfl_reason_disposition on (operator_code, reason_category, equipment_disposition, started_at DESC) — finance reporting (e.g., "all DUNNING_TERMINATION with WRITTEN_OFF equipment in May")

Sample data:

id	operator_code	subscription_id	reason_category	equipment_disposition	terminal_state	services_dispatched_count	services_succeeded_count
tfl_01HZ_KAMAU_MOVED	WIK	sub_01HZ_KAMAU_FIBER	CUSTOMER_REQUESTED	PENDING_COLLECTION	FULLY_REVOKED	2	2
tfl_01HZ_OTIENO_DUN_TERM	WIK	sub_01HZ_OTIENO_TRIPLEPLAY	DUNNING_TERMINATION	WRITTEN_OFF	FULLY_REVOKED	3	3
tfl_01HZ_FRAUD_TERM	WIK	sub_01HZ_FRAUD_CASE	ADMIN_DECISION	WRITTEN_OFF	FAILED_AND_ROLLED_BACK	2	1

termination_fulfillment_service_outcome

Column	Type	Notes
id	TEXT PK	ULID. Example: tfso_01HZ...
fulfillment_id	TEXT NOT NULL	FK to termination_fulfillment_ledger.id
service_ref	TEXT NOT NULL	Example: svc_VOIP_LINE
dispatch_order	INTEGER NOT NULL	1-based position in the dispatch sequence. Example: 1
provisioner_key	TEXT NULL	Resolved provisionerKey. Null if dispatch never reached Provisioner lookup. Example: VOIP_SIP_PROFILE
service_management_key	TEXT NULL	The key looked up in HomePass. Example: voice. Null for partner-fulfilled.

Column	Type	Notes
`target_node_code`	TEXT NULL	Example: `VOIPSWITCH-NRB-01`
`target_port`	TEXT NULL	Example: `EXT-5421`
`outcome`	TEXT NOT NULL	`SUCCESS` / `FAILURE` / `SKIPPED_NOT_ADDRESSABLE` / `SKIPPED_ABORTED_CHAIN` / `SUCCESS_THEN_COMPENSATED`
`failure_reason`	TEXT NULL	`NO_HOMEPASS_ENDPOINT_FOR_KEY` / `HOMEPASS_PORT_NOT_CAPTURED` / `UNKNOWN_PROVISIONER_KEY` / `ADAPTER_TIMEOUT` / `ADAPTER_ERROR`. Example: `ADAPTER_TIMEOUT`
`failure_detail`	TEXT NULL	Free-text detail. Example: `"VoipSwitch SOAP response timed out after 25000ms"`
`provisioner_outcome`	JSONB NULL	Full structured outcome returned by the Provisioner. Example: `{"success": true, "adapterRefs": {"deletedProfileId": "prof_voip_88421"}}`
`compensation_status`	TEXT NULL	`NOT_APPLICABLE` (default) / `COMPENSATED` / `COMPENSATION_FAILED`
`compensation_detail`	TEXT NULL	Free-text. Example: `"restoreService rollback succeeded in 1240ms"`
`started_at`	TIMESTAMP NOT NULL	Example: `2026-05-18 10:30:00.456`
`completed_at`	TIMESTAMP NULL	Example: `2026-05-18 10:30:02.301`
`duration_ms`	INTEGER NULL	Example: `1845`

Indexes:

- `idx_tfso_fulfillment` on `(fulfillment_id, dispatch_order)` — read service outcomes per attempt in order
- `idx_tfso_outcome` on `(outcome, completed_at DESC)` — failure investigation

Sample data (for `tfl_01HZ_OTIENO_DUN_TERM` — triple-play dunning termination, fully revoked):

id	fulfillment_id	service_ref	order	provisioner_key	service_mgmt_key	target_node_code	target_port	outcome
`tfso_01HZ_OTIENO_1`	`tfl_01HZ_OTIENO_DUN_TERM`	`svc_VOIP_LINE`	`1`	`VOIP_SIP_PROFILE`	`voice`	`VOIPSWITCH-NRB-01`	`EXT-5421`	`SUCCESS`
`tfso_01HZ_OTIENO_2`	`tfl_01HZ_OTIENO_DUN_TERM`	`svc_IPTV_BASIC`	`2`	`VERIMATRIX_IPTV`	`iptv_multicast`	`VERIMATRIX-VMC-NRB`	`STB_67890`	`SUCCESS`
`tfso_01HZ_OTIENO_3`	`tfl_01HZ_OTIENO_DUN_TERM`	`svc_INET_50M`	`3`	`HFC_DOCSIS_PROFILE`	`data`	`CMTS-NRB-WTL-01`	`1/2/8`	`SUCCESS`

`ful_05_config`

Per-operator config. One row per operator deployment.

Column	Type	Notes
`operator_code`	TEXT PK	Values: `WIK` / `WUG` / `WTZ` / `YASSN`
`default_partial_allowed`	BOOLEAN NOT NULL DEFAULT FALSE	Default for `partialAllowed`. `false` = abort-and-rollback on any Service failure. Example: `false`
`per_service_timeout_ms`	INTEGER NOT NULL DEFAULT 25000	Timeout per Provisioner method call. Example: `25000`
`overall_timeout_ms`	INTEGER NOT NULL DEFAULT 55000	Overall budget for one attempt including compensation. Example: `55000`
`cache_ttl_seconds`	INTEGER NOT NULL DEFAULT 300	TTL for Service / Package / HomePass read caches. Example: `300`
`updated_at`	TIMESTAMP NOT NULL	Example: `2026-05-18 09:00:00`
`updated_by`	TEXT NOT NULL	Example: `admin_user_42`

Sample data:

operator_code	partial_allowed	per_service_timeout_ms	overall_timeout_ms	cache_ttl_seconds
WIK	false	25000	55000	300
WUG	false	25000	55000	300
WTZ	false	25000	55000	300
YASSN	false	25000	55000	300

4. API surface

Execute termination fulfillment

```
POST /api/termination-fulfillment/execute HTTP/1.1
Authorization: Bearer <internal-svc-JWT>
Content-Type: application/json

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "customerId": "cust_01HZ_KAMAU",
  "operatorCode": "WIK",
  "packageRef": "pkg_INET_VOIP_RES",
  "homepassId": "hp_01HZ_KAREN_017",
  "reasonCategory": "CUSTOMER_REQUESTED",
  "equipmentDisposition": "PENDING_COLLECTION",
  "equipmentBindings": [
    {"equipmentRequirement": "GPON_ONT", "equipmentRef": "ont_01HZ_KAMAU"},
    {"equipmentRequirement": "ATA",      "equipmentRef": "ata_01HZ_KAMAU"}
  ],
  "fulfillmentCorrelationId": "saga-terminate-sub_01HZ_KAMAU-20260518T103000",
  "callerWorkflowKind": "SUB_WF_TERMINATE",
  "callerWorkflowRefId": "terminate_01HZ_KAMAU_MOVED",
  "policy": {
    "partialAllowed": false
  }
}
```

Response 200 (fully revoked):

```
{
  "fulfillmentId": "tfl_01HZ_KAMAU_MOVED",
  "terminalState": "FULLY_REVOKED",
  "reasonCategory": "CUSTOMER_REQUESTED",
  "equipmentDisposition": "PENDING_COLLECTION",
  "servicesDispatchedCount": 2,
  "servicesSucceededCount": 2,
  "services": [
    {
      "serviceRef": "svc_VOIP_LINE",
      "dispatchOrder": 1,
      "provisionerKey": "VOIP_SIP_PROFILE",
      "targetNodeCode": "VOIPSWITCH-NRB-01",
      "targetPort": "EXT-5421",
      "outcome": "SUCCESS",
      "provisionerOutcome": {"success": true, "adapterRefs": {"deletedProfileId": "prof_voip_88421"}}
    },
    {
      "serviceRef": "svc_INET_100M",
      "dispatchOrder": 2,
      "provisionerKey": "GPON_RES_PROFILE",
      "targetNodeCode": "OLT-NRB-WTL-01",
      "targetPort": "1/2/3",
      "outcome": "SUCCESS",
      "provisionerOutcome": {"success": true, "adapterRefs": {"deprovisionedPortAck": "OLT_PORT_FREE"}}
    }
  ],
  "completedAt": "2026-05-18T10:30:04.150Z",
  "durationMs": 4150
}
```

Partial failure with compensation:

```
→ 200 OK
{
  "fulfillmentId": "tfl_01HZ_FRAUD_TERM",
  "terminalState": "FAILED_AND_ROLLED_BACK",
  "reasonCategory": "ADMIN_DECISION",
  "services": [
    {
      "serviceRef": "svc_VOIP_LINE",
      "outcome": "SUCCESS_THEN_COMPENSATED",
      "compensationStatus": "COMPENSATED",
```

```
      "compensationDetail": "restoreService rollback succeeded in 1240ms"
    },
    {
      "serviceRef": "svc_INET_100M",
      "outcome": "FAILURE",
      "failureReason": "ADAPTER_TIMEOUT",
      "failureDetail": "Huawei NCE response timeout after 25000ms"
    }
  ]
}
```

Idempotency hit:

```
→ 200 OK
{
  "fulfillmentId": "tfl_01HZ_KAMAU_MOVED",
  "terminalState": "FULLY_REVOKED",
  "idempotentReturn": true
}
```

In-progress collision:

```
→ 409 Conflict
{
  "code": "FULFILLMENT_IN_PROGRESS",
  "fulfillmentId": "tfl_01HZ_KAMAU_MOVED_PRIOR",
  "startedAt": "2026-05-18T10:29:55Z"
}
```

Validation errors:

```
→ 400 Bad Request
{
  "code": "INVALID_EQUIPMENT_DISPOSITION",
  "violations": [
    {"field": "equipmentDisposition", "received": "RETURN_TO_VENDOR", "allowed": ["CUSTOMER_RETAINS","PENDING_COLLECTION",
  ]
}
```

Read a fulfillment

```
GET /api/termination-fulfillment/tfl_01HZ_KAMAU_MOVED HTTP/1.1
Authorization: Bearer <bo-agent-jwt>

→ 200 OK
{ ledger row + service-outcome rows nested }
```

Query

```
GET /api/termination-fulfillment?operator=WIK&subscriptionId=sub_01HZ_KAMAU_FIBER
GET /api/termination-fulfillment?operator=WIK&terminalState=FAILED_AND_ROLLED_BACK&since=2026-05-01
GET /api/termination-fulfillment?operator=WIK&reasonCategory=DUNNING_TERMINATION&equipmentDisposition=WRITTEN_OFF&since=20
GET /api/termination-fulfillment?operator=WIK&callerWorkflowRefId=terminate_01HZ_KAMAU_MOVED
```

5. Kafka events emitted

Topic prefix: `sophix.fulfillment.termination.*`

Event	Topic	Fires on	Consumers
`TerminationFulfillmentCompleted`	`sophix.fulfillment.termination.completed`	Terminal `FULLY_REVOKED`	SUB-WF-TERMINATE-01 (via Camunda message catch correlated on fulfillmentCorrelationId); analytics; OSR-RMA-01 EQP satellite trigger
`TerminationFulfillmentPartial`	`sophix.fulfillment.termination.partial`	Terminal `PARTIALLY_REVOKED`	SUB-WF-TERMINATE-01 + ICN-01 ops alerts
`TerminationFulfillmentFailed`	`sophix.fulfillment.termination.failed`	Terminal `FAILED_AND_ROLLED_BACK` / `FAILED_AT_FIRST_SERVICE` / `NOTHING_TO_REVOKE`	SUB-WF-TERMINATE-01 + ICN-01 ops alerts to `noc-team`
`CompensationFailed`	`sophix.fulfillment.termination.compensation-failed`	A compensation step failed	ICN-01 ops alerts to `noc-team` for manual cleanup

Sample `TerminationFulfillmentCompleted`:

```
{
  "eventType":   "TerminationFulfillmentCompleted",
  "aggregateId": "tfl_01HZ_KAMAU_MOVED",
  "timestamp":   "2026-05-18T10:30:04.150Z",
  "payload": {
    "operatorCode":            "WIK",
```

```
      "fulfillmentId":            "tfl_01HZ_KAMAU_MOVED",
      "subscriptionId":           "sub_01HZ_KAMAU_FIBER",
      "customerId":               "cust_01HZ_KAMAU",
      "reasonCategory":           "CUSTOMER_REQUESTED",
      "equipmentDisposition":     "PENDING_COLLECTION",
      "fulfillmentCorrelationId": "saga-terminate-sub_01HZ_KAMAU-20260518T103000",
      "callerWorkflowKind":       "SUB_WF_TERMINATE",
      "callerWorkflowRefId":      "terminate_01HZ_KAMAU_MOVED",
      "servicesDispatchedCount":  2,
      "servicesSucceededCount":   2,
      "services": [
        {"serviceRef":"svc_VOIP_LINE","provisionerKey":"VOIP_SIP_PROFILE","targetNodeCode":"VOIPSWITCH-NRB-01","targetPort":
        {"serviceRef":"svc_INET_100M","provisionerKey":"GPON_RES_PROFILE","targetNodeCode":"OLT-NRB-WTL-01","targetPort":"1/
      ],
      "completedAt": "2026-05-18T10:30:04.150Z",
      "durationMs":  4150
    }
  }
```


6. Worked example — Kamau moves to a no-coverage area

Kamau calls the Call Center to terminate his Wananchi Kenya fiber + voice subscription because he's relocating to a non-covered area. The CC agent verifies, captures `terminationTrigger=CUSTOMER_REQUESTED` + `terminationReasonCode=CUSTOMER_MOVED_NO_COVERAGE` per SUB-WF-TERMINATE-01. The agent confirms with Kamau that the ONT and ATA need to be returned (equipment disposition `PENDING_COLLECTION`). SUB-WF-TERMINATE-01 starts: validates → cancels any in-flight saga → puts state PENDING_TERMINATION → calls billing (`TERMINATION_INTENT`; computes final pro-rated invoice + closes wallet balance). Billing returns OK. SUB-WF-TERMINATE-01 reaches `callFulfillment`.

10:30:00: Worker calls:

```
POST /api/termination-fulfillment/execute
{"subscriptionId":"sub_01HZ_KAMAU_FIBER","customerId":"cust_01HZ_KAMAU","operatorCode":"WIK",
 "packageRef":"pkg_INET_VOIP_RES","homepassId":"hp_01HZ_KAREN_017",
 "reasonCategory":"CUSTOMER_REQUESTED","equipmentDisposition":"PENDING_COLLECTION",
 "equipmentBindings":[{"equipmentRequirement":"GPON_ONT","equipmentRef":"ont_01HZ_KAMAU"},
                      {"equipmentRequirement":"ATA","equipmentRef":"ata_01HZ_KAMAU"}],
 "fulfillmentCorrelationId":"saga-terminate-sub_01HZ_KAMAU-20260518T103000",
 "callerWorkflowKind":"SUB_WF_TERMINATE","callerWorkflowRefId":"terminate_01HZ_KAMAU_MOVED"}
→ 200 fulfillmentId=tfl_01HZ_KAMAU_MOVED, terminalState=IN_PROGRESS
```

10:30:00 (internal): FUL-05 inserts `termination_fulfillment_ledger` row. Loads Package `pkg_INET_VOIP_RES` → Services `[svc_INET_100M, svc_VOIP_LINE]`. Both `is_addressable=true` and `consumption_model=CONTINUOUS`. Loads HomePass hp_01HZ_KAREN_017, reads `service_management_endpoints = {"data": {"nodeCode":"OLT-NRB-WTL-01","port":"1/2/3"}, "voice": {"nodeCode":"VOIPSWITCH-NRB-01","port":"EXT-5421"}}`. Applies reverse order per R-OR-1 (no explicit policy): voice first, then data.

10:30:00.250: Service 1 (reverse-ordered) `svc_VOIP_LINE`, key `voice` → target `(VOIPSWITCH-NRB-01, EXT-5421)`. Provisioner `VoipSipProvisioner.revokeService("CUSTOMER_REQUESTED", "PENDING_COLLECTION", ...)`. Adapter pushes a delete-profile command to VoipSwitch via side channel. Returns `{success:true, adapterRefs:{deletedProfileId:"prof_voip_88421"}}`.

`termination_fulfillment_service_outcome` row 1: outcome=SUCCESS, dispatch_order=1, duration_ms=1200.

10:30:01.450: Service 2 `svc_INET_100M`, key `data` → target `(OLT-NRB-WTL-01, 1/2/3)`. Provisioner `GponResProvisioner.revokeService("CUSTOMER_REQUESTED", "PENDING_COLLECTION", ...)`. Adapter sends an OLT-port deprovision command via Huawei NCE GLDS tunnel. Returns `{success:true, adapterRefs:{deprovisionedPortAck:"OLT_PORT_FREE"}}`.

`termination_fulfillment_service_outcome` row 2: outcome=SUCCESS, dispatch_order=2, duration_ms=2700.

10:30:04.150: FUL-05 aggregates: 2 SUCCESS / 0 FAILED → terminal `FULLY_REVOKED`. Ledger updated. Kafka emits `TerminationFulfillmentCompleted`.

10:30:05: SUB-WF-TERMINATE-01's Camunda message catch fires on the Kafka event. The saga commits: closes Kamau's pause history if any active, puts master fields (`terminated_at=2026-05-18T10:30:05Z`, `termination_trigger=CUSTOMER_REQUESTED`, `termination_reason_code=CUSTOMER_MOVED_NO_COVERAGE`, `prior_status_at_termination=ACTIVE`), puts state `TERMINATED`, emits `SubscriptionTerminated`.

10:30:06+: Downstream — OSR-RMA-01 EQP satellite subscribes to `SubscriptionTerminated` events with `equipmentBindings.length > 0` AND `equipmentDisposition=PENDING_COLLECTION` and creates a pickup work order for the two equipment refs. NOT-01 sends Kamau a confirmation SMS + email with the pickup window. BIL-02 finalizes any closing invoice line.

Kamau's network access is gone — his ONT can no longer authenticate to the OLT, his ATA can no longer register with VoipSwitch. The physical equipment sits at his address until the contractor agent visits within the next 5 working days.

7. Module ecosystem

Role	Module	What it does
Owner	`fulfillment` module (FUL-05 component)	Owns the orchestration logic, the ledger tables, the Provisioner adapter dispatch, the Kafka emit
Upstream caller	SUB-WF-TERMINATE-01	Synchronous REST to `POST /api/termination-fulfillment/execute`
Read dependencies	SIP-01 (Package), PLM-CFG-01 (Service), RLM-CFG-01 (HomePass)	Read-only catalog lookups
Adapter callees	Provisioner adapter beans (`GponResProvisioner`, `HfcDocsisProvisioner`, `VoipSipProvisioner`, `VerimatrixProvisioner`, ...)	In-module code beans; not separate DDs
Downstream consumers	SUB-WF-TERMINATE-01 (via Camunda message catch on Kafka), OSR-RMA-01 EQP satellite (subscribes to `SubscriptionTerminated` emitted downstream after FUL-05 success), ICN-01 (ops alerting), EM-04 (reporting)	Consume Kafka events

8. Dependencies

Dependency	Owner	Status
SUB-WF-TERMINATE-01 (caller)	subscription team	✓ v0.9
SIP-01 Package Management	plm team	✓ v1.0
PLM-CFG-01 Service Catalog	plm team	✓ v0.8 (Provisioner interface defined here)
RLM-CFG-01 HomePass Configuration	network/plm team	✓ v1.0 (service_management_endpoints structure defined here)
SUB-LM-01 Subscription Lifecycle Management	subscription team	✓ v1.0
FUL-03 Service Activation (sibling — shares Provisioner registry)	fulfillment team	✓ v1.0
FUL-04 Restriction & Non-Payment Suspension Fulfillment (sibling)	fulfillment team	✓ v1.0
OSR-RMA-01 EQP satellite (downstream consumer for equipment pickup)	inventory team	✓ v1.0
FOUNDATION_DB, FOUNDATION_KAFKA, FOUNDATION_CACHE, FOUNDATION_AUTH	infra team	✓
Provisioner adapter implementations — new methods `revokeService` + `restoreService` (in-module code, not DD)	fulfillment team	■ to be implemented as part of this module
PLM-CFG-01 v0.9 — Provisioner interface extension to add `revokeService` / `restoreService` methods (alongside FUL-04's additions)	plm team	■ captured as open coordination item below

B — Current TZ

Status: To be filled in after codebase cartography review.

Workshop materials (Deep dive 05 Provisioning + Service Termination flow, 2026-04-30) note that the legacy termination chain reaches AAA / NMS / Verimatrix / VoipSwitch via the Supercontroller, and that "service revoked end-to-end via Supercontroller → NMS" is the legacy pattern. Specific table names, command sequences, and per-vendor termination behavior in the existing codebase require source-code review to document accurately.

Until that cartography lands, this DD does not assert specific legacy structural claims about termination handling. The §C section below describes V3 design intent without depending on unverified details.

C — V3 design intent

FUL-05 is new functionality in V3. The V3 model:

- **Single inbound contract** for service termination. SUB-WF-TERMINATE-01 calls one endpoint; FUL-05 fans out to Services internally via the Provisioner registry.
- **Provisioner adapter dispatch parity with FUL-03 / FUL-04**. Same registry, same per-vendor adapter beans, same target resolution from HomePass `service_management_endpoints`. The adapters gain a new interface method (`revokeService`) plus a compensation companion (`restoreService` for in-attempt rollback only).
- **Structured** `equipmentDisposition` propagated to the Provisioner. Vendor adapters can implement different cleanup behavior for `CUSTOMER_RETAINS` (preserve AAA record, disable only) vs `PENDING_COLLECTION` / `WRITTEN_OFF` (full clean-out).
- **Ledger + per-service-outcome audit**. Every termination attempt produces a durable audit row plus one row per Service revoked. Retention indefinite.
- **Reverse-default ordering**. When no explicit policy supplied, services are revoked in reverse of the Package's declared order. Voice + TV before data — minimizes brief windows of "partial service" that could surface as support calls.
- **Idempotency by** `(subscriptionId, fulfillmentCorrelationId)` — saga retries are safe.
- **Equipment retrieval cleanly delegated** to OSR-RMA-01 EQP satellite via the `SubscriptionTerminated` event. FUL-05 owns network state; OSR owns physical equipment.

Migration approach

Migration of any in-flight terminations from legacy source data to V3 is deferred to a dedicated migration design document produced after codebase cartography. The V3 module deploys with empty ledger tables and begins recording from cutover.

1. Deploy `fulfillment` module with the `termination_fulfillment_ledger` + `termination_fulfillment_service_outcome` tables empty.
2. Implement Provisioner adapter methods (`revokeService`, `restoreService`) on each existing adapter bean. Same vendor wrappers as FUL-03 / FUL-04.
3. Per-operator seed of `ful_05_config`.
4. SUB-WF-TERMINATE-01 v0.9 expects FUL-05 as a saga step — switching from the placeholder "no-op" step to the real FUL-05 endpoint is a saga configuration change.

Open coordination items

- **PLM-CFG-01 v0.9 Provisioner interface extension**: bundled with FUL-04's additions. The complete v0.9 interface adds `applyRestriction`, `liftRestriction`, `suspendService`, `resumeService` (from FUL-04) plus `revokeService` and `restoreService` (from this DD). Plus the `targetPort` parameter previously captured by FUL-03. PLM team owns the formalization.
- **Provisioner code components** — `GponResProvisioner`, `HfcDocsisProvisioner`, `VoipSipProvisioner`, `VerimatrixProvisioner`, plus any partner-fulfilled adapters — each gains two new methods (`revokeService`, `restoreService`). The methods are in-module code, not DDs.
- `equipmentDisposition` **semantic per vendor** — each Provisioner adapter interprets the disposition value:
- `CUSTOMER_RETAINS`: disable AAA record but preserve it for legal-retention reasons; do not delete OLT port profile (rare; operator-specific policy).
- `PENDING_COLLECTION`: delete AAA profile, deprovision OLT port, shutdown vendor service. Equipment is expected to come back to the warehouse.
- `WRITTEN_OFF`: same as `PENDING_COLLECTION` from the network plane (equipment value is being accepted as a loss; no special vendor behavior). Distinguishes for downstream finance / reporting.

Workshop deep-dive on per-vendor command sets remains pending; per-operator policy on `CUSTOMER_RETAINS` is captured in the Provisioner adapter code, not in FUL-05 config.

- **Downstream OSR-RMA-01 EQP satellite contract** — OSR-RMA-01 EQP subscribes to `SubscriptionTerminated` (emitted by SUB-WF-TERMINATE-01 after FUL-05 returns success). The subscription criterion: `equipmentBindings.length > 0` AND `equipmentDisposition IN ('PENDING_COLLECTION')`. `WRITTEN_OFF` bindings do NOT trigger a pickup WO; the equipment is logged as a write-off via the inventory system. `CUSTOMER_RETAINS` bindings also skip pickup. This subscription contract is captured here for traceability and will land formally in OSR-RMA-01 v1.1 when the EQP satellite incorporates the FUL-05 event source.